INTERNSHIP PROJECT REPORT

RetailPulse: AI-Powered Customer Analytics & Demand Forecasting

Submitted To :

Zidio Development

Data Science & Analytics Internship

Team Members & Contributions:

Name	Key Responsibilities
Shania Shamshi	**• Data Collection • Data Cleaning & Preprocessing • Data Preparation • Exploratory Data Analysis (EDA) • Business Insights Generation • Feature Engineering**
Nitesh Kumar Mishra	**• Machine Learning Model Development • Deep Learning (LSTM) Implementation • Customer Segmentation • Demand Forecasting • Churn Prediction • Best Algorithm Selection using Model Comparison • Hyperparameter Tuning & Model Evaluation • Final Trained Models**
Abhay Sharma	**• Streamlit Dashboard Development • Dashboard UI/UX Design • Machine Learning Model Integration • Interactive Visualizations & Analytics • Deployment (Streamlit Cloud / Render / AWS) • GitHub Repository Management • Final Testing & README Documentation • Live Demo Preparation**

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1. Abstract:

RetailPulse: AI-Powered Customer Analytics & Demand Forecasting is a comprehensive data analytics and machine learning project designed to help businesses make data-driven decisions by analyzing customer behavior, forecasting product demand, and improving inventory planning. The project integrates data preprocessing, exploratory data analysis (EDA), feature engineering, machine learning, deep learning, and interactive visualization into a unified analytics platform.

The project begins with collecting and preparing retail transaction data through extensive data cleaning and preprocessing to ensure accuracy and consistency. Exploratory Data Analysis (EDA) is performed to uncover customer purchasing patterns, sales trends, product performance, and key business insights. Feature engineering techniques are applied to create meaningful variables that enhance predictive model performance.

Multiple machine learning algorithms are developed and evaluated to perform demand forecasting and customer churn prediction. In addition, a Long Short-Term Memory (LSTM) deep learning model is implemented to improve forecasting accuracy by capturing temporal patterns in sales data. Customer segmentation techniques are also employed to identify distinct customer groups and support personalized business strategies. Hyperparameter tuning and model comparison are conducted to select the most effective predictive models.

To make these insights accessible, an interactive Streamlit dashboard is developed, providing real-time visualizations, predictive analytics, and business intelligence through an intuitive user interface. The application is deployed on a cloud platform and managed through GitHub for collaborative development and version control.

Overall, RetailPulse demonstrates how artificial intelligence, machine learning, and business analytics can be integrated to transform raw retail data into actionable insights, enabling organizations to optimize inventory, improve customer retention, enhance operational efficiency, and support strategic decision-making.

Keywords: Customer Analytics, Demand Forecasting, Machine Learning, Deep Learning, LSTM, Customer Segmentation, Churn Prediction, Streamlit, Data Analytics, Business Intelligence, Feature Engineering, Predictive Analytics.

2. Objectives:

The primary objective of this project is to develop an AI-powered analytics platform that enables businesses to analyze customer behavior, forecast product demand, and support data-driven decision-making. The project aims to transform raw retail transaction data into meaningful business insights through data analytics, machine learning, and interactive visualization.

The specific objectives of this project are:

1. To collect, clean, and preprocess retail transaction data to ensure data quality and consistency.
2. To perform Exploratory Data Analysis (EDA) to identify customer purchasing behavior, sales trends, and product performance.
3. To engineer meaningful features that improve the performance of predictive machine learning models.
4. To develop and compare multiple machine learning models for demand forecasting and customer churn prediction.
5. To implement a Long Short-Term Memory (LSTM) deep learning model for accurate time-series demand forecasting.
6. To perform customer segmentation for identifying distinct customer groups and supporting personalized business strategies.
7. To evaluate and optimize predictive models using model comparison and hyperparameter tuning techniques.
8. To design and develop an interactive Streamlit dashboard for real-time visualization of business insights and predictive analytics.
9. To deploy the application on a cloud platform and manage the project using GitHub for collaboration and version control.
10. To provide actionable insights that help businesses optimize inventory management, improve customer retention, increase operational efficiency, and support strategic decision-making.

3.Problem Statement

Retail businesses generate large volumes of transactional data every day; however, much of this data remains underutilized for strategic decision-making. Many organizations rely on manual analysis and historical observations for inventory planning, customer management, and sales forecasting. This often leads to inaccurate demand estimation, stock shortages of high-demand products, excess inventory of slow-moving items, and difficulty in identifying customers who are likely to stop purchasing.

In addition, the absence of an integrated analytics platform makes it challenging for business users to interpret complex datasets and derive actionable insights. Traditional spreadsheet-based analysis is time-consuming, error-prone, and unable to provide predictive intelligence required for modern retail operations.

To address these challenges, the **RetailPulse: AI-Powered Customer Analytics & Demand Forecasting** project was developed. The project leverages data analytics, machine learning, and deep learning techniques to transform raw retail transaction data into meaningful business insights. It performs comprehensive data cleaning, feature engineering, exploratory data analysis, customer segmentation, churn prediction, and demand forecasting. Furthermore, an interactive **Streamlit** dashboard was developed to visualize business insights and predictive results in a user-friendly manner.

The proposed solution enables businesses to better understand customer behavior, forecast future demand, optimize inventory management, identify at-risk customers, and support data-driven decision-making, ultimately improving operational efficiency and customer satisfaction.

4.Technologies Used:

The RetailPulse project was developed using a combination of data analytics, machine learning, visualization, and web application technologies. Each technology played a specific role in data processing, predictive modeling, and dashboard development.

Technology	Purpose
Python	Core programming language used for data analysis, machine learning, and application development.
Pandas	Data cleaning, preprocessing, manipulation, and analysis of retail transaction data.
NumPy	Numerical computations and array operations.
Matplotlib	Creation of statistical and analytical visualizations.
Seaborn	Advanced data visualization and exploratory data analysis (EDA).
Scikit-learn	Machine learning model development, customer segmentation, churn prediction, model evaluation, and hyperparameter tuning.
TensorFlow	Development and training of the LSTM deep learning model for demand forecasting.
Streamlit	Development of an interactive web-based analytics dashboard.
Git	Version control and collaborative development.
GitHub	Source code management, collaboration, and project hosting.
Jupyter Notebook	Data exploration, experimentation, preprocessing, and model development.
Visual Studio Code (VS Code)	Source code development and application programming.

5.Data Cleaning & Preprocessing:

To ensure high-quality and reliable analysis, the raw retail transaction dataset underwent a comprehensive data cleaning and preprocessing process. The following steps were performed before conducting exploratory data analysis and building predictive models.

Data Cleaning Steps

1. Date Format Conversion

The **InvoiceDate** column was converted into the DateTime format to enable accurate time-based analysis and feature extraction.

2. Missing Value Handling

Records containing missing values in the **Description** and **Customer ID** columns were removed to maintain data consistency and ensure reliable customer-level analysis.

3. Duplicate Record Removal

Duplicate transactions were identified and removed to eliminate redundant information and improve data quality.

4. Invalid Quantity Removal

Transactions with **Quantity ≤ 0** were excluded, as they do not represent valid product purchases.

5. Invalid Price Removal

Records with **Price ≤ 0** were removed because products with zero or negative prices are considered invalid for sales analysis.

6. Cancelled Order Removal

Invoices beginning with the letter **'C'** were identified as cancelled transactions and excluded from the dataset to ensure that only completed sales were analyzed.

Outcome

After completing the data cleaning process, the dataset became more accurate, consistent, and suitable for exploratory data analysis, feature engineering, machine learning model development, and demand forecasting. The cleaned dataset formed the foundation for all subsequent stages of the RetailPulse analytics pipeline.

6. Dataset Information:

The project utilizes the Online Retail II dataset, which contains transactional records of a UK-based online retail company. The dataset was collected over a two-year period and was further processed through data cleaning, preprocessing, and feature engineering to prepare it for exploratory data analysis and predictive modeling.

Attribute	Value
Original Records:	**1,067,371**
Original Features:	**8**
Processed Records:	**779,425**
Processed Features:	**19**
Time Period:	**December 2009 – December 2011**

7.Feature Engineering:

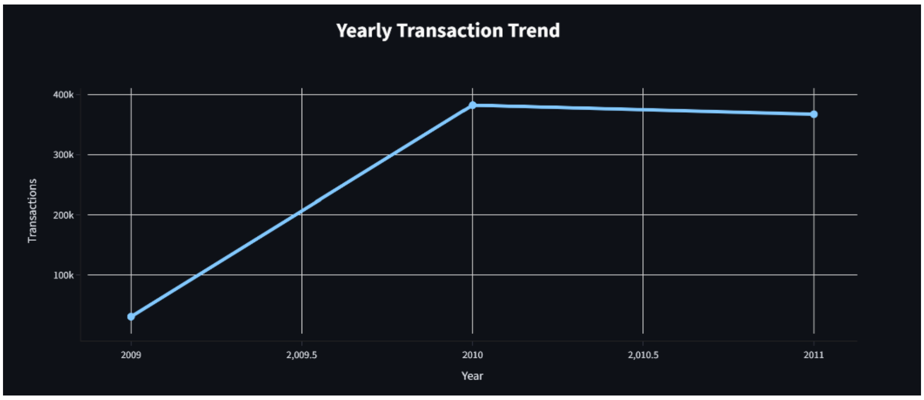
Feature Engineering is the process of creating new variables (features) from the existing dataset to improve data analysis, uncover hidden patterns, and enhance the performance of machine learning models. In the RetailPulse project, several time-based and business-oriented features were derived from the cleaned retail transaction dataset.

Feature	Description
Revenue:	Total revenue generated per transaction, calculated as **Quantity × Price**.
Year:	Year extracted from the transaction date for yearly trend analysis.
Month:	Numeric month extracted from the transaction date.
MonthName:	Month name for improved dashboard visualization and reporting.
Day:	Day of the month extracted from the transaction date.
DayOfWeek:	Name of the weekday used for weekly sales analysis.
Hour:	Hour of the transaction used to identify peak shopping hours.
Quarter:	Quarter of the year (Q1–Q4) for quarterly business analysis.
IsWeekend:	Binary feature indicating whether the transaction occurred on a weekend (1) or weekday (0).
BasketSize:	Total number of items purchased within a single invoice.
OrderValue:	Total monetary value of each customer order (invoice).

8.Exploratory Data Analysis (EDA):

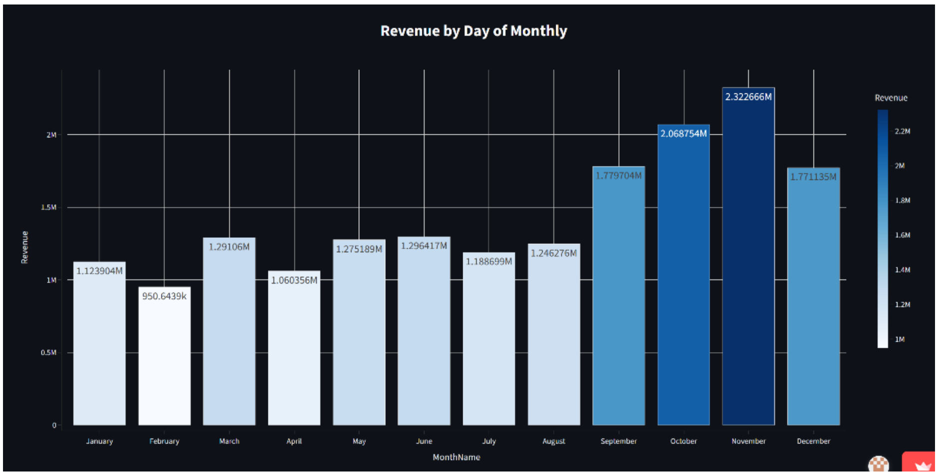
The objective of this notebook is to explore the cleaned retail dataset, identify important patterns and trends, understand customer purchasing behavior, analyze product and country-wise sales, and generate business insights for further feature engineering and predictive modeling.

Figure 8.1: Yearly Transaction Sales



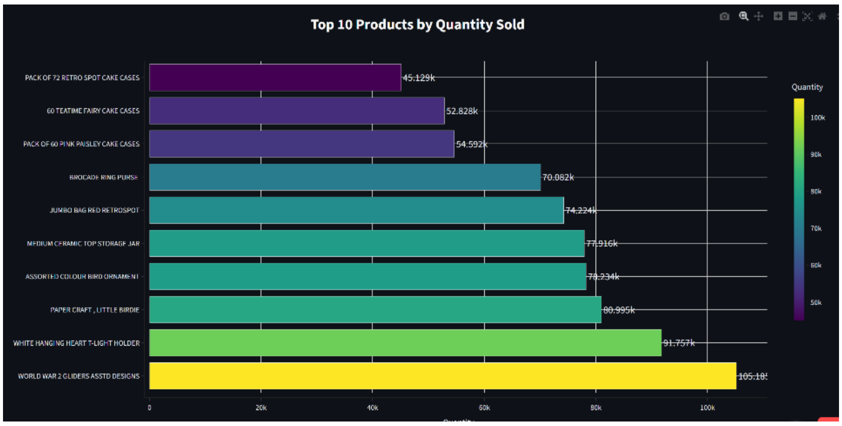
Business Insight: The consistent increase in transaction volume indicates strong business growth and increasing customer engagement. The slight decline in 2011 may suggest seasonal fluctuations or changing customer demand. These insights can help businesses evaluate long-term performance and improve inventory planning.

Figure 8.2: Monthly Revenue Trend



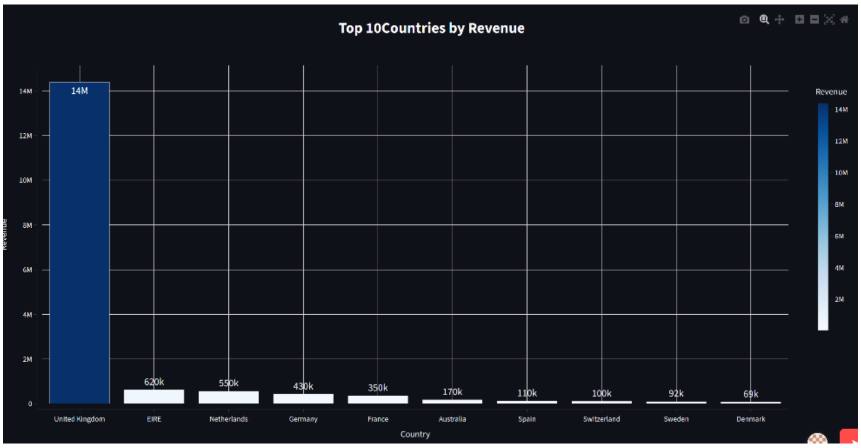
Business Insight: November recorded the highest revenue (~2.32M), making the final quarter the peak sales period.February had the lowest revenue (~950K), indicating seasonal decline. Revenue was stable from January–August and increased significantly from September onward.

Figure 8.3: Top 10 products by quantity sold



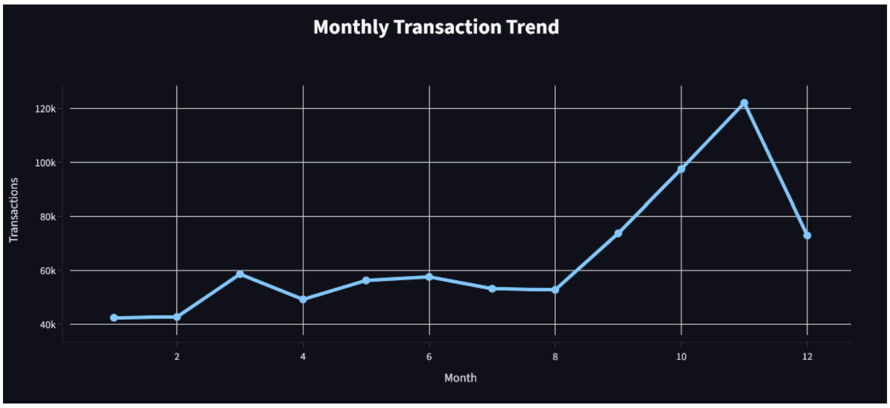
Business Insight: WORLD WAR 2 GLIDERS ASSTD DESIGNS is the best-selling product with 105,185 units sold.WHITE HANGING HEART T-LIGHT HOLDER is the second most sold product.The Top 10 products have very high sales, showing they are the most popular items.These products should be kept well-stocked and promoted to maximize sales**.**

Figure 8.4: Top 10 Countries by Revenue



Business Insight: United Kingdom generated the highest sales revenue (14.39 million), far exceeding all other countries.EIRE and Netherlands are the second and third highest revenue-generating countries.Most of the revenue comes from a few countries, with the United Kingdom dominating sales.The company should maintain its strong presence in the UK while expanding in other high-potential markets

Figure 8.5: Monthly Transactions Trend

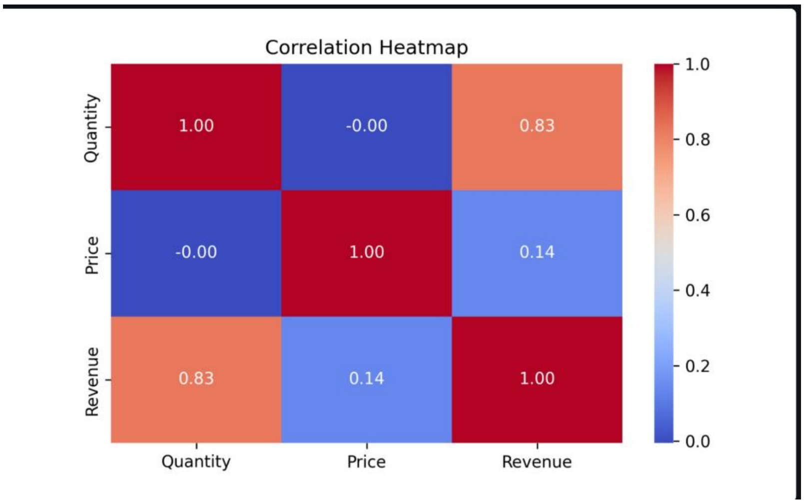


Business Insight : Transaction volume generally increased toward the end of each year.November 2011 recorded the highest number of transactions (63,168), indicating peak business activity.Transaction counts were comparatively lower at the beginning of the year and highest during the festive/holiday season (October-November)

Figure 8.6: Top 10 Customers by Orders

Business Insight : Customer 14911 placed the highest number of orders (398), followed by 12748 (336).A small group of customers places orders very frequently, indicating a strong base of loyal and repeat buyers.Retaining these high-value customers through loyalty programs and personalized offers can improve long-term revenue

Figure 8.7: Correlation Heatmap



Business Insight : Quantity and Revenue have a strong positive correlation (0.83), indicating that higher quantities sold generally result in higher revenue.Unit Price and Revenue have a weak positive correlation (0.14), suggesting that revenue is influenced more by sales volume than product price.Quantity and Unit Price have almost no correlation (-0.005), showing that the quantity purchased is largely independent of product price.

9. Machine Learning Methodology

9.1 Data Preparation :

Before training the machine learning models, the processed dataset was further prepared to ensure accurate predictions. Relevant features were selected, customer-level information was aggregated, and additional variables required for predictive modeling were created.

9.2 RFM Feature Creation :

RFM (Recency, Frequency, and Monetary) analysis was performed to summarize customer purchasing behavior.

Feature	Description
Recency:	Number of days since the customer's last purchase
Frequency:	Total number of unique purchases
Monetary:	Total amount spent by the customer

	CustomerID	Recency	Frequency	Monetary
0	12346.0	9	9	203.50
1	12358.0	55	1	1429.83
2	12359.0	46	2	838.89
3	12361.0	5	1	109.20
4	12362.0	61	1	130.00

9.3 Customer Churn Prediction : The objective was to identify customers who are likely to stop purchasing so that businesses can take proactive customer retention measures.

Algorithm Used

- **Random Forest Classifier**

Output Files

Output	Description
customer_churn_prediction.csv	Churn prediction results
churn_model.pkl	Trained Random Forest model

	count
Churn	
0	1661
1	912

9.4 Customer Segmentation : The objective of customer segmentation was to group customers with similar purchasing behavior, enabling businesses to develop targeted marketing strategies and improve customer relationship management.

Algorithm Used

- **K-Means Clustering**

Output Files

Output	Description
customer_segmentation.csv	Customer segment assignments
customer_segmentation.pkl	Trained K-Means model

9.5 Demand Forecasting : The objective was to forecast future product demand, helping businesses optimize inventory planning and reduce stockout or overstock situations.

Two forecasting approaches were implemented.

A. Machine Learning Forecasting

Machine learning models were used to predict future product demand based on historical sales patterns.

Output Files

Output	Description
forecast_predictions.csv	Forecasted demand values
forecast_model.pkl	Trained forecasting model

B. Deep Learning Forecasting

An **LSTM (Long Short-Term Memory)** neural network was implemented to capture sequential sales patterns and improve forecasting accuracy.

Output Files

Output	Description
lstm_predictions.csv	LSTM forecast results
lstm_demand_forecasting.keras	Trained LSTM model

9.4 Model Comparison : Multiple machine learning algorithms were evaluated to identify the most suitable model for each prediction task.

Output	Description
model_comparison.csv	Performance comparison of all models

9.5 Hyperparameter Tuning : Hyperparameter tuning was performed to optimize the performance of the Random Forest model using **GridSearchCV**.

Best Parameters:

Max Depth: 3

Min Samples Split: 2

Number of Estimators: 50

Best Accuracy : 100% (1.00)

Output Files

Output	Description
hyperparameter_report.csv	Hyperparameter tuning results
best_churn_model.pkl	Optimized Random Forest model

9.6 Best Models

Customer Segmentation: **K-Means Clustering**

Customer Churn Prediction: **Random Forest Classifier**

Demand Forecasting: **LSTM Neural Network**

9.7 Final Trained Models

The final trained models generated during the project are summarized below:

Model File	Purpose
customer_segmentation.pkl:	**Customer Segmentation**
churn_model.pkl:	**Customer Churn Prediction**
best_churn_model.pkl:	**Optimized Churn Prediction Model**
forecast_model.pkl:	**Machine Learning Demand Forecasting**
lstm_demand_forecasting.keras:	**Deep Learning Demand Forecasting**

10. Streamlit Dashboard

10.1 Overview

An interactive web application was developed using **Streamlit** to present customer analytics, demand forecasting, and inventory insights in a user-friendly interface. The dashboard enables users to explore sales trends, customer behavior, predictive analytics, and business insights without requiring technical knowledge.

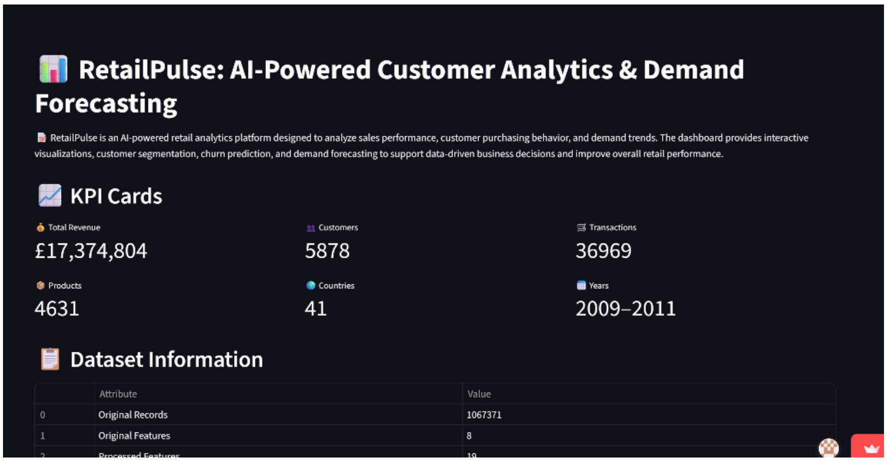
10.2 Dashboard Features

The Streamlit application provides the following functionalities:

- Interactive sales and revenue analysis
- Customer segmentation visualization
- Customer churn prediction
- Demand forecasting using Machine Learning and LSTM
- Product and inventory insights
- Dynamic charts and business analytics
- User-friendly navigation and responsive interface

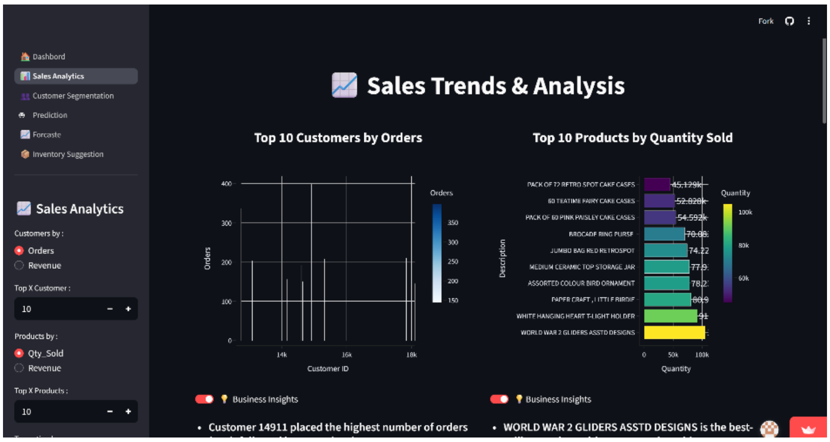
10.3 Dashboard Screens

Figure 10.1: Home Dashboard



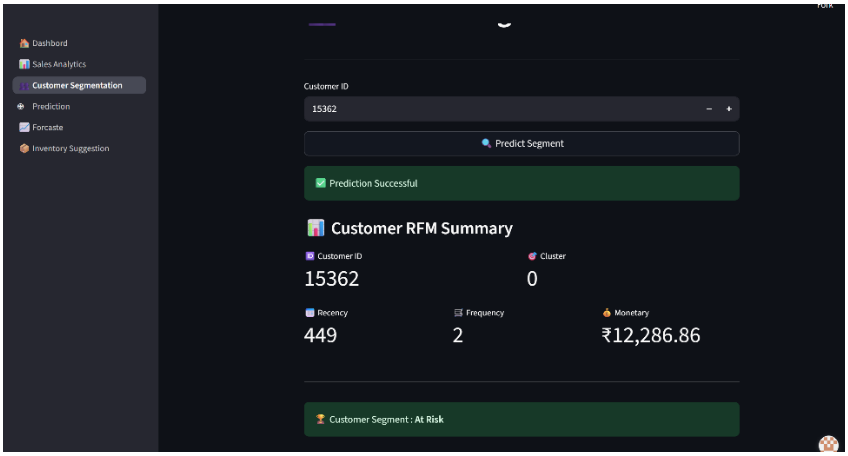
The home page provides an overview of the RetailPulse platform and allows users to navigate to different analytics modules

Figure 10.2: Sales Analytics Dashboard



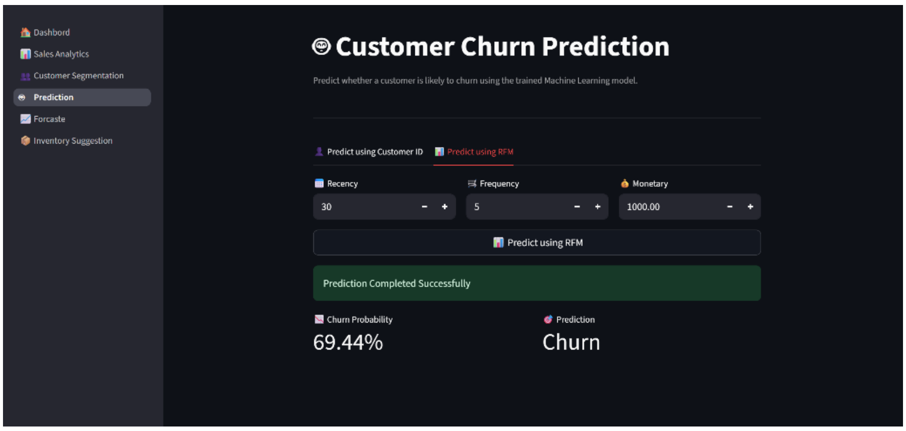
Displays customer insights, purchasing behavior, customer segments, and churn-related analytics through interactive visualizations.

Figure 10.3: Customer Segmentation Dashboard



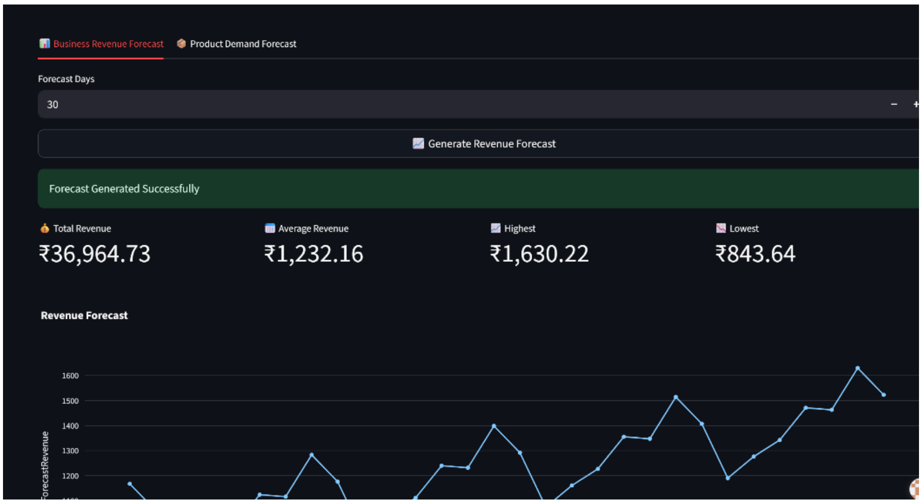
The Customer Segmentation dashboard classifies customers into different segments using the K-Means Clustering algorithm based on their Recency, Frequency, and Monetary (RFM) values. Users can enter a Customer ID to predict the customer's segment.

Figure 10.4: Prediction Results



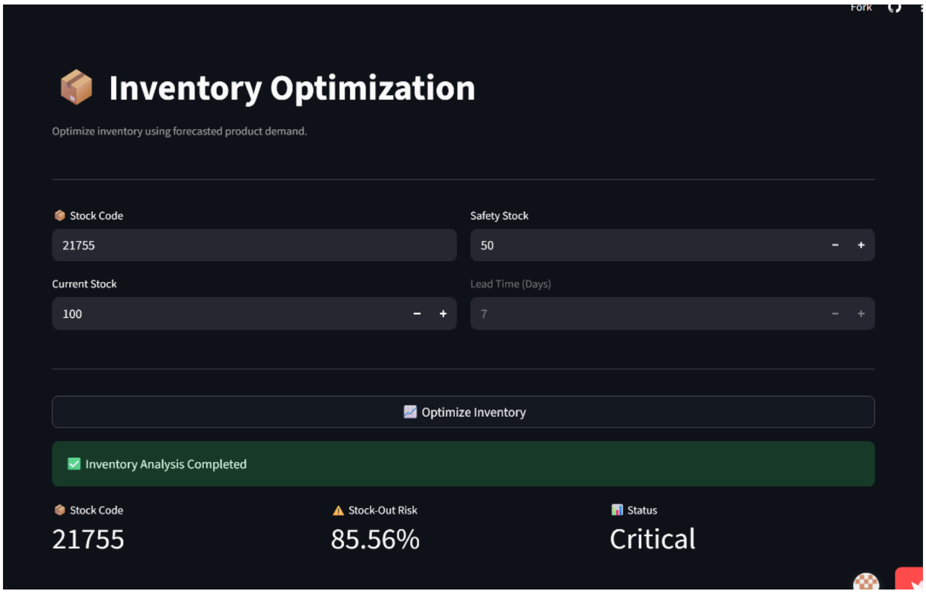
Displays the prediction results generated by the trained machine learning models for customer churn.

Figure 10.5: Demand Forecasting Dashboard



Shows predicted product demand generated using Machine Learning and LSTM models, helping businesses optimize inventory planning.

Figure 10.6: Inventory Sugeestion Dashboard



Provides insights into stock availability, high-demand products, and inventory performance to support informed decision-making.

Deployment

The RetailPulse web application was deployed using **Streamlit Community Cloud**, providing a publicly accessible and interactive platform for customer analytics, demand forecasting, and inventory intelligence. The deployment enables users to access the application through a web browser without requiring local installation.

Deployment Details

Platform	Purpose
Streamlit Community Cloud:	Web Application Deployment
GitHub:	Source Code Hosting and Version Control

- **Live Application:** https://retail-pulse-zd.streamlit.app/
- **GitHub Repository:** https://github.com/Abhay936/Retail-Pulse

11. Results & Discussion

The RetailPulse project successfully transformed raw retail transaction data into meaningful business insights through data analytics, machine learning, and interactive visualization. The major outcomes of the project are summarized below:

- Successfully cleaned and preprocessed **1,067,371** retail transaction records, resulting in **779,425** high-quality records for analysis.
- Expanded the dataset from **8 original features** to **19 processed features** using feature engineering techniques.
- Performed Exploratory Data Analysis (EDA) to identify sales trends, customer purchasing behavior, product performance, and seasonal demand patterns.
- Developed a **Customer Segmentation** model using the **K-Means Clustering** algorithm to group customers based on purchasing behavior.
- Built a **Customer Churn Prediction** model using the **Random Forest Classifier** to identify customers at risk of leaving.
- Implemented **Demand Forecasting** using both Machine Learning techniques and an **LSTM Deep Learning** model for future sales prediction.
- Compared multiple machine learning algorithms and selected the best-performing models based on evaluation metrics.
- Optimized the Random Forest model using **GridSearchCV** for improved predictive performance.
- Developed and deployed an interactive **Streamlit** dashboard to visualize customer analytics, demand forecasts, and inventory insights.
- Successfully hosted the application on **Streamlit Community Cloud**, making it accessible through a web browser.

Discussion

The project demonstrates how artificial intelligence and data analytics can improve retail decision-making. By combining data preprocessing, exploratory analysis, predictive modeling, and interactive visualization, the solution enables businesses to forecast demand, identify customer segments, detect churn risk, and optimize inventory planning. The deployed dashboard makes these insights easily accessible to non-technical users, supporting data-driven business decisions.

12. Conclusion

The **RetailPulse: AI-Powered Customer Analytics & Demand Forecasting** project successfully integrated data analytics, machine learning, and deep learning techniques to solve key retail business challenges. Starting from raw transactional data, the project involved data cleaning, feature engineering, exploratory data analysis, predictive modeling, and dashboard development to generate meaningful business insights.

The implemented solution provides customer segmentation, churn prediction, and demand forecasting capabilities that support inventory optimization, customer retention, and strategic planning. The interactive Streamlit dashboard further enhances usability by presenting analytics and predictions through a simple and intuitive interface.

Overall, RetailPulse demonstrates how modern AI-powered analytics can help businesses improve operational efficiency, reduce inventory-related risks, and make informed, data-driven decisions.

13. References

The following resources were used during the development of this project:

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